

C02-AT1-ANALYTICAL PROBLEM SOLVING

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COURSE : Object Oriented Analysis and Design
CODE/NAME CSA1104

SLOT: B

1, Use Case Diagram - Hospital Management

a, Actors

- * Patient
- * Doctor
- * Administrator

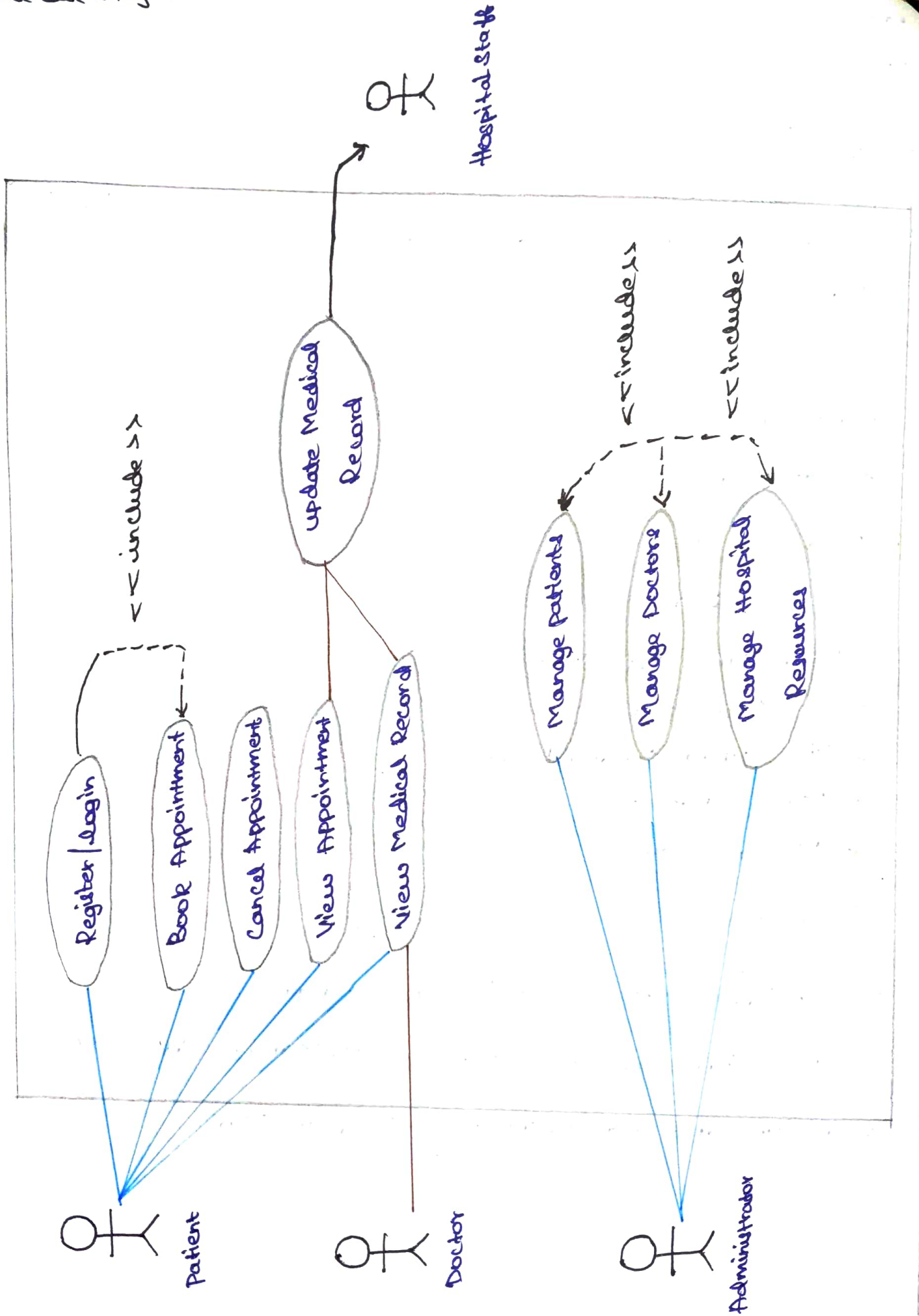
b, Main Use Case

- * Register / Login
- * Book / Cancel Appointment
- * View Appointment
- * View / update Medical Record
- * Manage Patients
- * Manage Doctors

c, Relationships

- * Book Appointment → <<include>> Login
- * Cancel Appointment → <<extend>> Manage Appointment
- * Doctor and Administrator → Hospital staff

d, Use Case Diagram



2, Class Diagram - Library Management System

a, Main classes

- * Book : bookId, title, author, availability
- * Member : memberId, name, email
- * Librarian : librarianId, name
- * IssueRecord : issueId, issueDate, returnDate

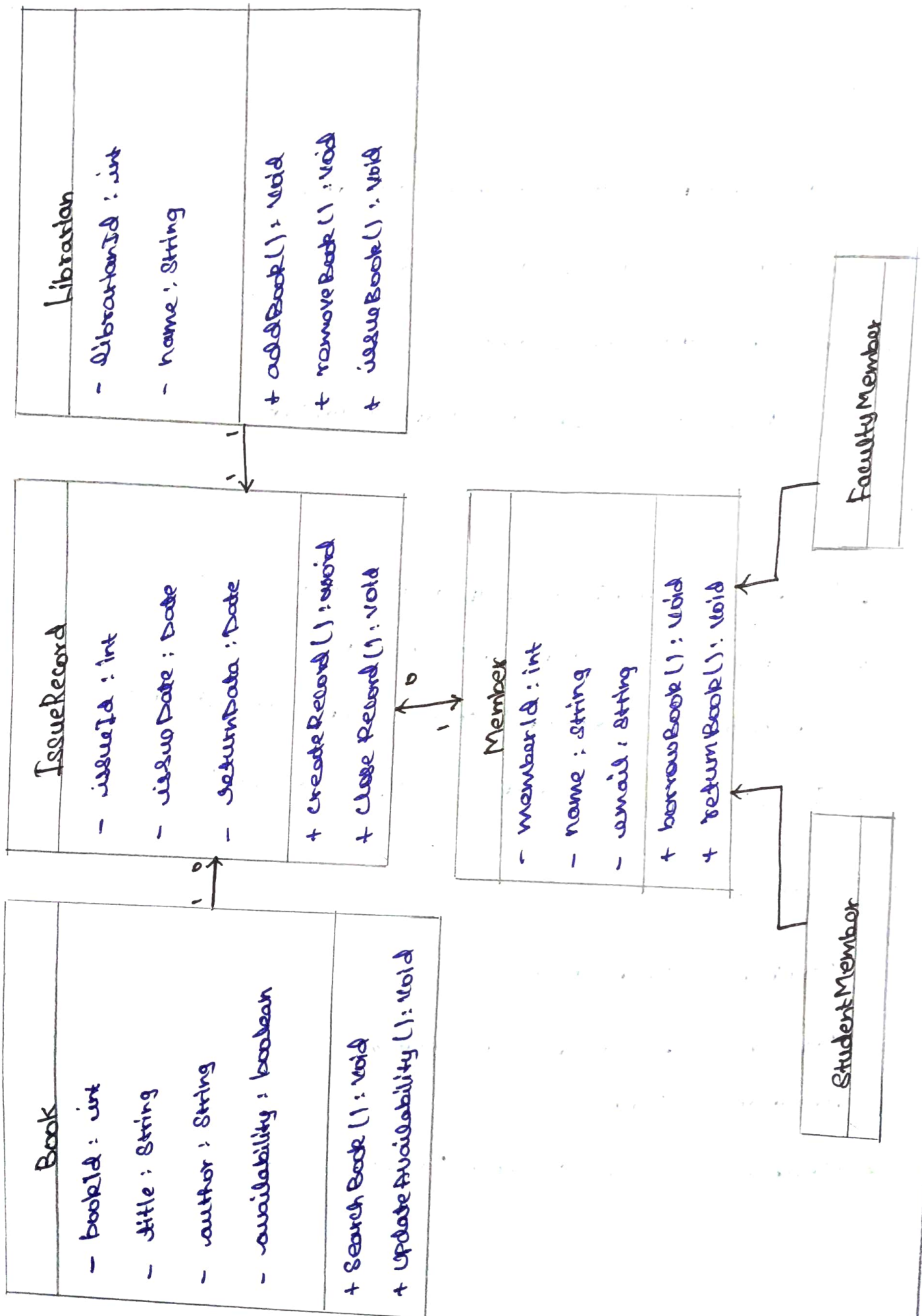
b, Methods

- Book → SearchBook(), updateAvailability()
- Member → borrowBook(), returnBook()
- Librarian → addBook(), removeBook(), issueBook()
- IssueRecord → CreateRecord(), closeRecord()

c, Relationships

- * Member → Book : Association
- * Member → IssueRecord : Association
- * Librarian → Book : Manager
- * Member → Student Member / faculty Member

4. Class Diagram



3, Sequence Diagram - Online Food Ordering System

a, Objects

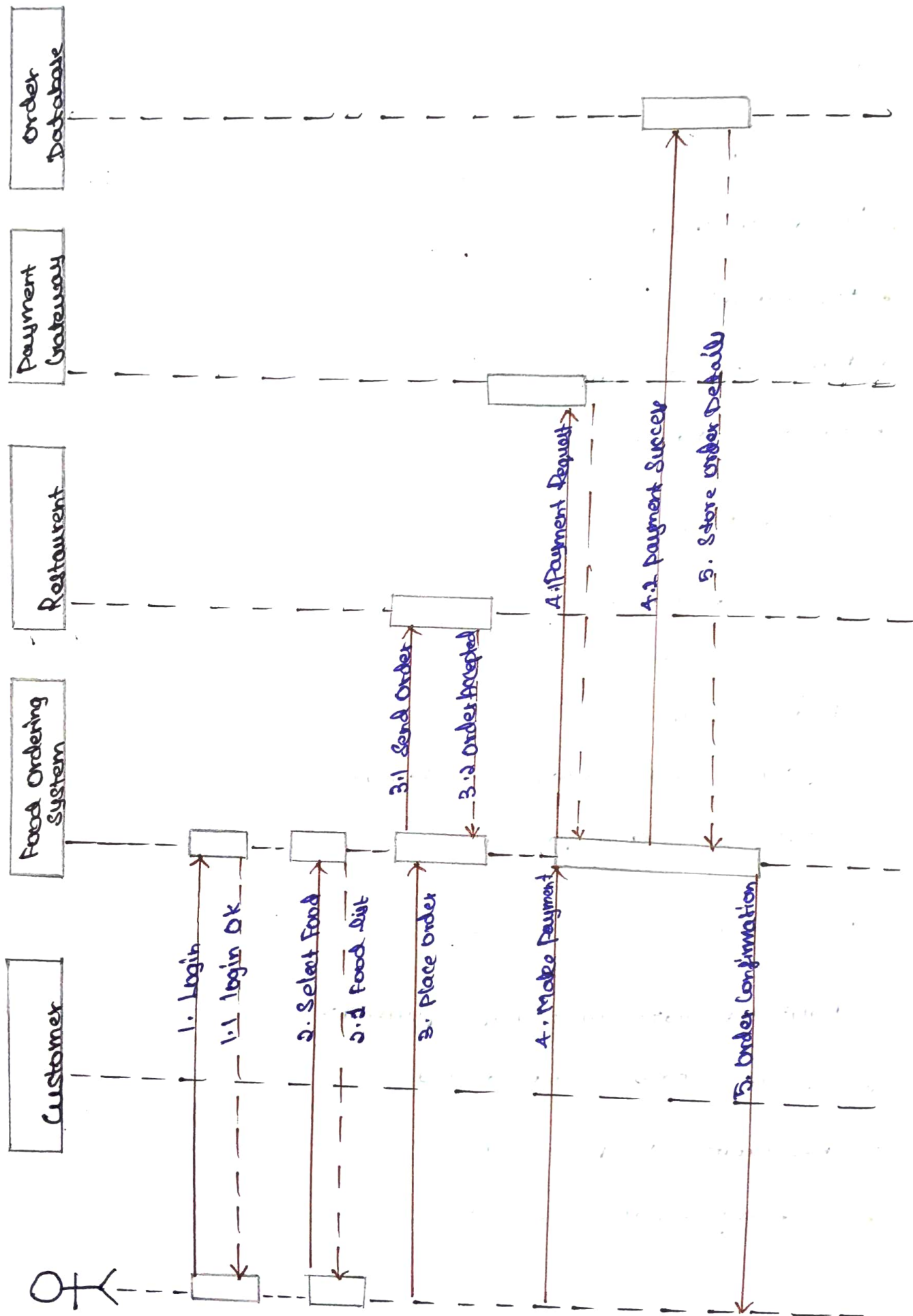
- * Customer
- * Food Ordering System
- * Restaurant
- * Payment Gateway
- * Order Database

b, Message Flow

- Customer logs in
- Selects food
- Places order
- System sends order to restaurant
- Customer makes payment

c, Processing

- * Payment Gateway verifies payment
- * System stores order details
- * Restaurant accepts order
- * System sends Confirmation



Activity Diagram - Student Admission System

a, Activities

- * Start
- * Submit Application
- * Upload Documents
- * Verify Documents
- * Check Eligibility
- * Pay Fees
- * Confirm Enrolment
- * End

b, Decision Nodes

- * Document Valid
- * Student eligible
- * Payment successful

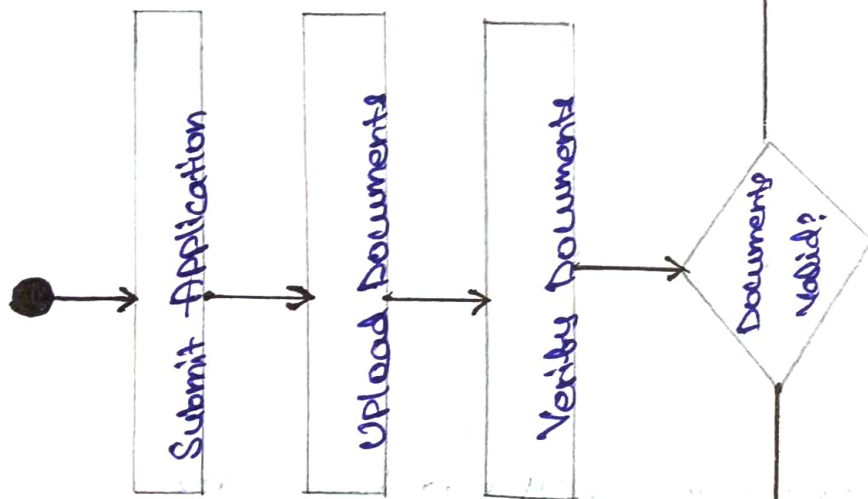
c, Flow

Start → Submit Application → upload Documents
→ Verify Documents

* If invalid → Reject Application → End

* If Valid → Check Eligibility

* If eligible → Pay Fee → confirm Enrolment
→ End



Reject Application

